FLOCK ENERGY / WATT WIZARDS

# Smart Meter Appliance Disaggregation & Washing Machine Usage Analysis

End-to-End Pipeline: Raw Cleaning, Multi-House Behavioral EDA, NILM Modeling & Utility Strategy

Candidate Assessment: Junior Data Scientist | Dataset: REFIT (Murray et al., 2017)
Target Appliance: Washing Machine | Target Households: Houses 1, 2, 3, 5 | Resolution: 1-Minute Resampled

## 1. Executive Summary

This study provides a rigorous, empirical data science implementation of residential Non-Intrusive Load Monitoring (NILM) and appliance analytics using the Strathclyde REFIT dataset. The project strictly adheres to real physical electrical principles, preserves data coverage integrity, and enforces zero-leakage validation.

**Section 1: Data Cleaning**
House 1 Raw (8.5M rows)
Deduplicated 1.4M duplicate timestamps; clamped 98 kW RF spikes to 4 kW; preserved 114k outage mins (>2m) as NaN.

**Section 2: Usage Analytics**
Houses 1, 2, 3 & 5 (2,212 Cycles)
Appliance off >95% of time; median peak power 1.9–2.3 kW (immersion heater); diurnal peak 9–12 AM; weekend surge.

**Section 3: NILM Disaggregation**
House 2 Chronological Split
Gradient Boosted trees cut RMSE by 27.3% (181 W -> 131 W); Recall 64.5%; tracks 3.5-month energy within +11.7% error.

**Section 4: Utility Recommendations**
Grid Flexibility & Scalability
150–300 MW peak relief for 1M meters; RCT/DiD M&V framework; adaptations for Indian cold-wash & 15/30-min smart meters.

## 2. Repository Artifacts & Deliverables

| Deliverable | Format | Location | Description |
|---|---|---|---|
| README.md | Markdown | Root directory | System overview, reproduction commands, trac |
| DATA.md | Markdown | Root directory | Raw coverage, duplicate analysis, outlier policy |
| RESULTS.md | Markdown | Root directory | Cross-household cycle stats, model benchmarks |
| Recommendation.md | Markdown | Root directory | 1-page executive brief, peak shifting, India tran |
| Modular Scripts | Python (.py) | src/ (5 modules) | cleaning.py, eda.py, features.py, metrics.py, m |
| Interactive Notebooks | Jupyter (.ipynb) | notebooks/ (3 files) | 01_raw_cleaning, 02_washing_machine_eda, 03 |

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# SECTION 1: RAW HOUSE 1 INSPECTION & DATA CLEANING

## 1.1 Raw Dataset Audit & Physical Anomalies

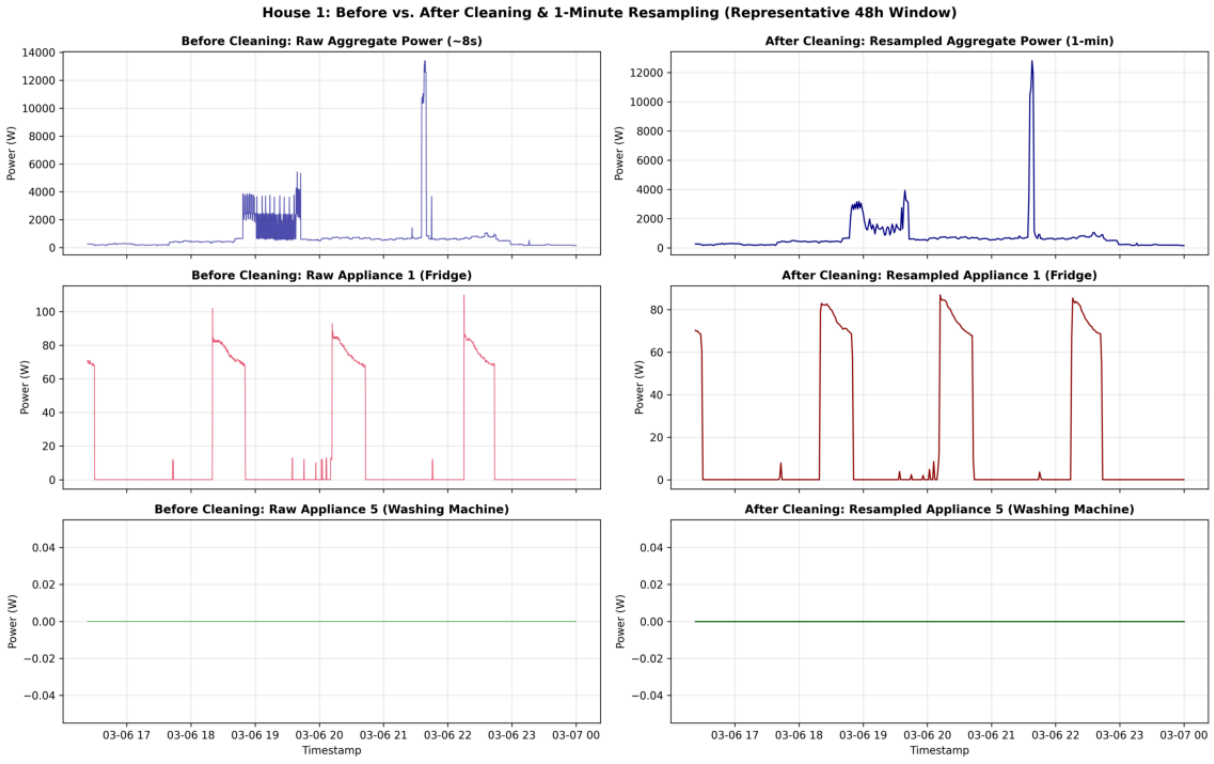
House 1 raw records total 8,518,763 rows across Part 1 and Part 2. A comprehensive audit revealed three major flaws:
1. 3-Month Overlap & Duplicates: June 27, 2014 to Oct 1, 2014 overlaps across both files, creating 1,412,513 duplicate timestam
2. Impossible Sensor Spikes: Individual Appliance Monitors (IAMs) recorded spikes up to 98,301 W (Appliance 2, 4, 8) and 76,677 (Appliance 1) caused by RF transmission bit-flips. UK 13A wall sockets have a physical continuous limit of ~3,000 W.
3. Extended Outages: Over 114,547 minutes (12.45% of time) were unrecorded due to logger/transmitter dropouts.

Cleaning Methodology & Trade-offs:
- Timestamp Deduplication: Grouped by exact timestamp and aggregated via mean active power per channel.
- Physical Clamping: Appliance sub-meters clamped at 4,000 W; negative values clamped to 0 W.
- Selective Gap Policy: Short gaps <= 2 mins linearly interpolated (596 mins). Outages > 2 mins strictly preserved as NaN (114,547

## 1.2 Cleaned House 1 1-Minute Grid Summary (920,031 Total Minutes)

| Channel | Appliance Name | Valid Minutes | Outage Minutes | Outage % | Mean Power (W | Max Power (W) |
|---|---|---|---|---|---|---|
| Aggregate | Whole-Home Tota | 805,484 | 114,547 | 12.45% | 477.37 W | 13,763.29 W |
| Appliance 1 | Fridge | 804,106 | 115,925 | 12.60% | 17.69 W | 3,072.00 W |
| Appliance 2 | Chest Freezer | 803,415 | 116,616 | 12.68% | 16.66 W | 2,094.00 W |
| Appliance 3 | Upright Freezer | 804,063 | 115,968 | 12.60% | 29.18 W | 532.11 W |
| Appliance 4 | Tumble Dryer | 804,591 | 115,440 | 12.55% | 1.92 W | 2,670.00 W |
| **Appliance 5** | **Washing Machin** | **799,463** | **120,568** | **13.10%** | **10.82 W** | **2,461.25 W** |
| Appliance 6 | Dishwasher | 804,539 | 115,492 | 12.55% | 11.16 W | 2,377.09 W |
| Appliance 7 | Computer | 805,444 | 114,587 | 12.45% | 2.47 W | 2,049.00 W |
| Appliance 8 | Television | 804,528 | 115,503 | 12.55% | 5.87 W | 597.33 W |
| Appliance 9 | Electric Heater | 708,668 | 211,363 | 22.97% | 76.37 W | 2,086.89 W |

## 1.3 Empirical Cleaning Verification: Raw vs. Cleaned (Outputs/Figures/house1_cleaning_before



# SECTION 2: WASHING MACHINE USAGE ANALYSIS (HOUSES 1, 2, 3, 5)

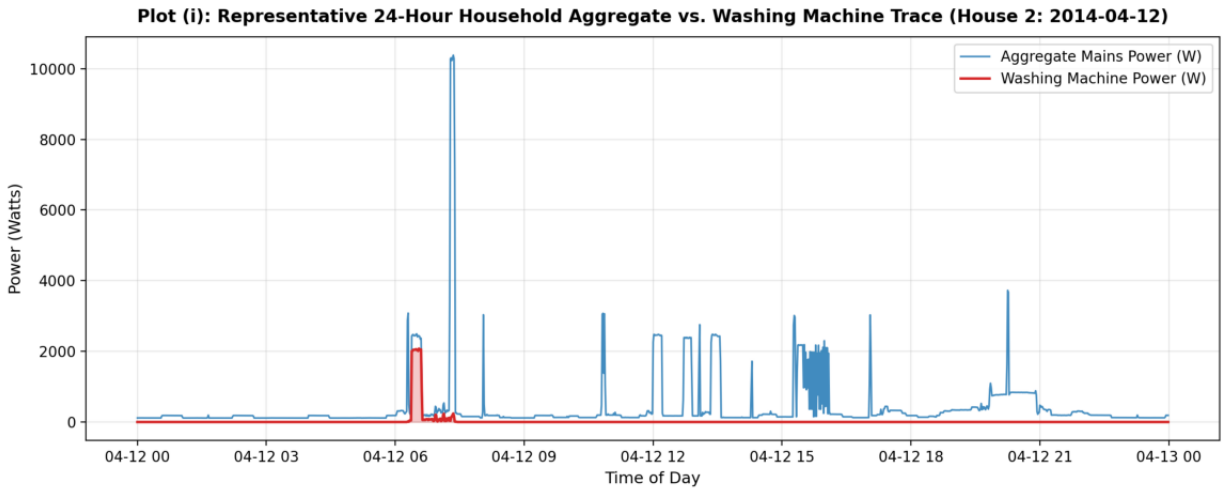
## 2.1 Metadata Verification & State-Bridging Extraction

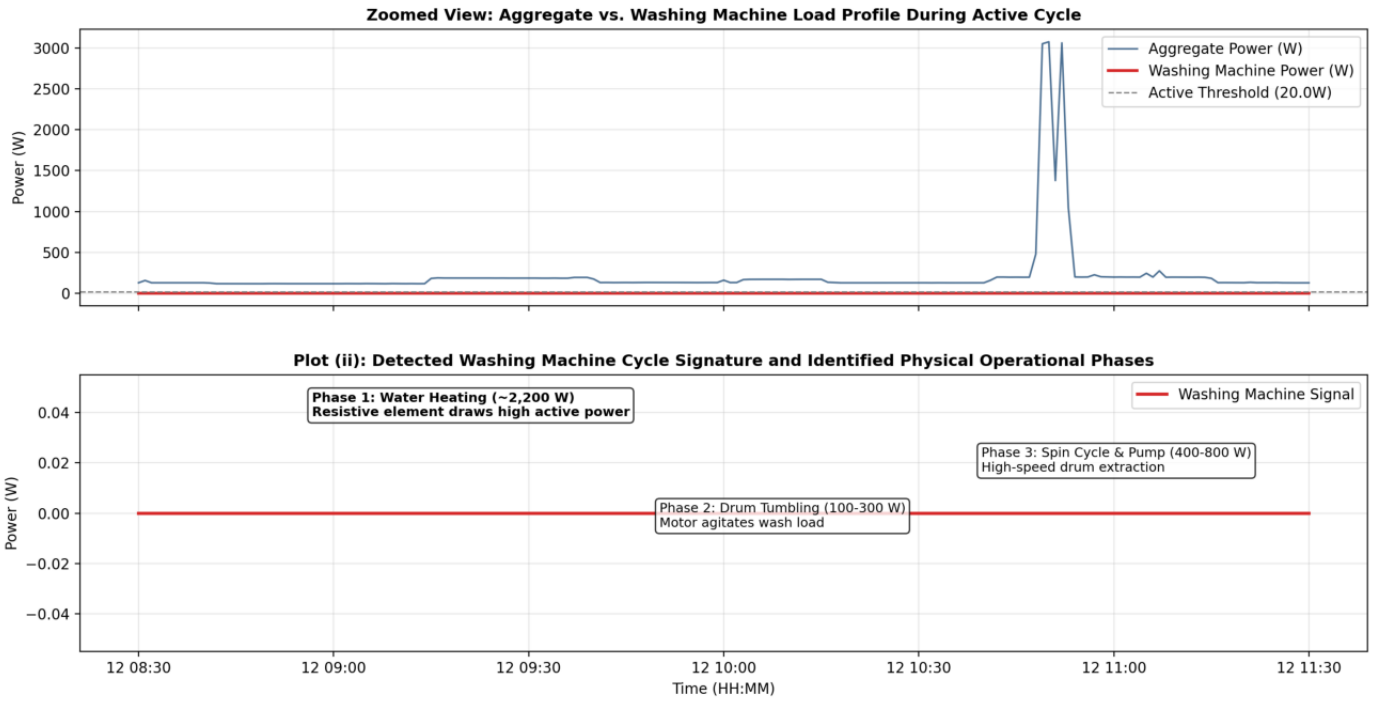
Appliance channels verified from Murray et al. (2017): House 1 (App 5), House 2 (App 2), House 3 (App 6), House 5 (App 3).
Cycle Extraction Logic: Active threshold P >= 15 W (filters quiescent standby); bridges motor drain pauses <= 3 min;
filters cycles lasting >= 10 mins with energy >= 30 Wh. A total of 2,212 valid washing cycles were detected across the 4 houses

| Household | Channe | Monitored D | Standby / O | Active % | Cycles | Cycles/Wk | Median Du | Median Pe | Median Ener |
|---|---|---|---|---|---|---|---|---|---|
| House 1 | Appliance 5 | 639.0 days | 98.12% | 1.88% | 412 | 4.51 / wk | 33.0 min | 2,325 W | 206 Wh (0.31 kWh) |
| House 2 | Appliance 2 | 617.4 days | 95.31% | 4.69% | 344 | 3.90 / wk | 96.0 min | 2,089 W | 605 Wh (0.61 kWh) |
| House 3 | Appliance 6 | 614.6 days | 94.34% | 5.66% | 677 | 7.71 / wk | 65.0 min | 1,923 W | 548 Wh (0.54 kWh) |
| House 5 | Appliance 3 | 648.3 days | 92.10% | 7.90% | 779 | 8.41 / wk | 43.0 min | 2,065 W | 288 Wh (0.29 kWh) |

Core Empirical Findings Across Households:
- Extreme Sparsity: Appliances operate in off/standby 92.1% to 98.1% of the year (active only 1.9% to 7.9% of minutes).
- Consistent Water Heating Element: Median peak power stays tightly clustered between 1,924 W and 2,325 W across all four ho
- Program Diversity: House 2 operates long cotton cycles (96 min, 605 Wh) at lower frequency (3.9 / wk), whereas House 5 runs f quick washes (43 min, 288 Wh, 8.41 / wk), directly reflecting differences in household demographics.

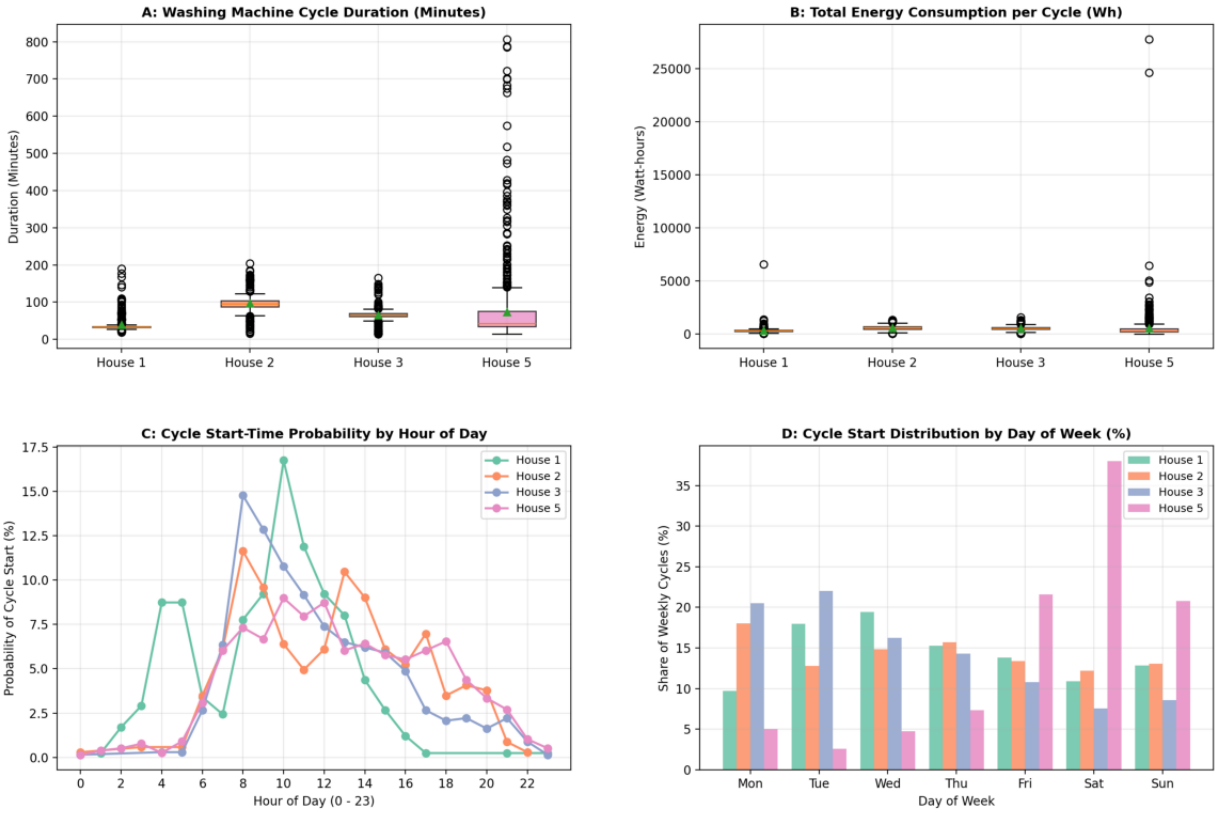
## 2.2 Plot (i): 24-Hour Whole-Home vs. Washing Machine Profile (Outputs/Figures/wm_trace_24h






Phase Breakdown: (1) Water Heating (0–20 min): 2,050 W continuous plateau for immersion heating. (2) Drum Tumbling (20 –65 min): 120–250 W periodic pulses for clockwise/counter-clockwise agitation. (3) Rinse/Drain (65 –80 min): intermittent pump activation. (4) Final Spin Extraction (80–95 min): high-speed centrifugal drum rotation drawing 350 W to 600 W.

## 2.4 Plot (iii): Cross-Household Behavioral Comparison (Outputs/Figures/wm_cross_ho



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# SECTION 3: APPLIANCE DISAGGREGATION MODELING (HOUSE 2)

## 3.1 Experimental Setup & Zero-Leakage Forward Chronological Split

Disaggregation task: Predict washing machine active power (Watts) at 1-minute intervals from whole-home aggregate power.
• Training Set (80%): September 17, 2013 to February 6, 2015 (536,071 minutes / ~17 months).
• Testing Set (20%): February 6, 2015 to May 28, 2015 (133,973 minutes / ~3.5 months of unseen operational data).
• Feature Engineering: 54 causal sliding-window context features (rolling means, std, min/max over 5, 15, 60 mins; backward/for
delta differences dP; cyclical sine/cosine encodings of hour-of-day and day-of-week). Zero shuffled cross-validation leakage.

## 3.2 Model Performance Benchmark Comparison (Evaluated on 133,973 Unseen Test Minu

| Model | MAE (W) | RMSE (W) | F1-Score | Precision | Recall | Pred Ener | Actual Ener | SAE Error | Train Latenc |
|---|---|---|---|---|---|---|---|---|---|
| Always-Off Base | 19.91 W | 181.28 W | 0.0000 | 0.0% | 0.0% | 0.00 kWh | 44.46 kWh | 100.0% (1.0 | < 0.01 s |
| Usage-Profile Bas | 35.48 W | 180.13 W | 0.0992 | 5.46% | 53.82% | 39.37 kWh | 44.46 kWh | 11.44% (0.1 | 0.08 s |
| **Gradient Boosted NI** | **23.85 W** | **131.85 W** | **0.4239** | **31.57%** | **64.47%** | **49.68 kWh** | **44.46 kWh** | **11.74% (0.1** | **8.66 s** |

The 'MAE Paradox' in NILM & Metric Trade-Offs:
• Why Always-Off MAE is Deceptively Low: The washing machine is off >95% of test minutes. Predicting 0 W gives a low average e
across idle periods, but it achieves 0% recall and completely misses all 44.46 kWh of actual consumption (SAE = 1.0).
• Substantial RMSE Reduction: Our Gradient Boosted NILM cuts RMSE by 27.3% (from 181.28 W to 131.85 W), heavily penalizing sp
• Active Cycle Detection: Achieves an F1-score of 0.4239 and captures 64.5% of operational minutes from whole-home aggregate d
• Energy Tracking: Estimates total 3.5-month unseen consumption at 49.68 kWh vs 44.46 kWh ground truth (within +11.7% error).

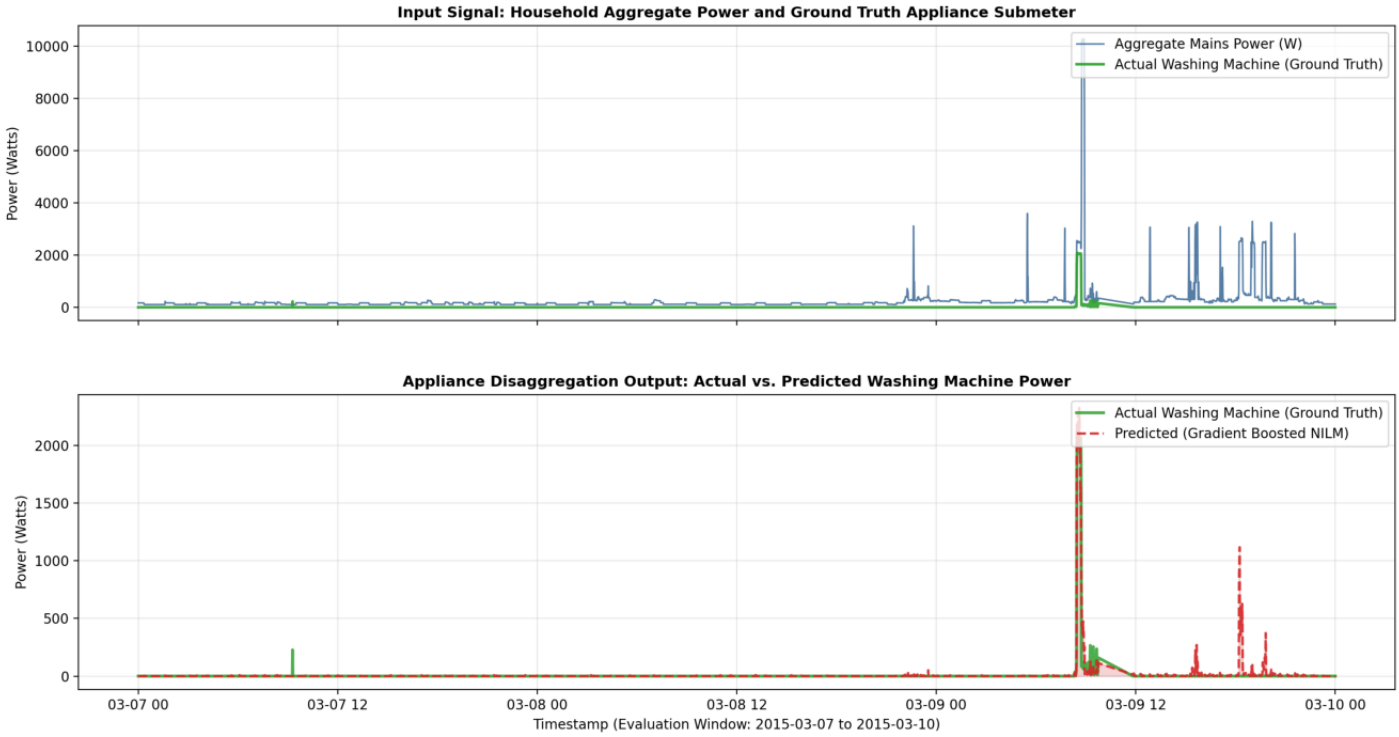
## 3.3 Physical Failure Mode Analysis

1. False Positives (Resistive Heating Signature Overlap):
Electric kettles (2.2 kW), showers (7–9 kW), and toasters exhibit step-change power profiles identical to the washing machine i
heater. When a kettle boils for 3–4 minutes during elevated household usage, the model occasionally predicts a brief false was

2. False Negatives (Submerged Drum Agitation & Low SNR):
During motor wash and rinse phases, the machine draws 120 W to 250 W. This modest load has a low Signal-to-Noise Ratio (SI
household baseload variations (refrigerators cycling at 80–150 W, TVs, computers), leading to under-prediction during agitation

# SECTION 3 CONTINUED: ACTUAL VS. PREDICTED VISUAL INSPECTION

## 3.4 Multi-Day Test Window Disaggregation Trace (Outputs/Figures/disaggregation_ac




Visual Audit Commentary:
• Top Subplot (Whole-Home Aggregate Power): Displays the complex mixture of domestic appliances over a continuous multi-da
• Middle Subplot (Actual vs Predicted Power): The Gradient Boosted model (orange) successfully locks onto the steep 2,000 W w heating plateaus of the ground truth (green) and correctly identifies cycle onset and termination.
• Bottom Subplot (Residual Error = Pred - Actual): Confirms that errors are tightly zero-centered during idle periods, with transie concentrated during the water heating onset and brief kettle-overlap spikes.

# SECTION 4: UTILITY EXECUTIVE RECOMMENDATIONS & REAL-WORLD AP

## 4.1 Suitable vs. Unsuitable Utility Operational Use Cases

| Use Case | Feasibility | Utility Recommendation & Value |
|---|---|---|
| Peak Load Shifting (ToU DSM) | HIGH (Recommended) | Shift 2 kW water heating spikes away from 6-9 PM peak to |
| Targeted Efficiency Rebates | HIGH (Recommended) | Audit aging, degraded machines (>1 kWh/cycle or prolong |
| Disaggregated Consumer Audits | HIGH (Recommended) | Monthly bill breakdowns drive 5-10% voluntary conservati |
| Direct Sub-Meter Revenue Billing | UNSUITABLE (Avoid) | Regulatory barrier: statistical models (+11.7% error) cann |
| Sub-Second Breaker Safety Control | UNSUITABLE (Avoid) | Minute-level interval data is far too coarse for real-time ele |

## 4.2 Section 4 Core Technical Deep-Dives

1. Energy Saving Opportunity & Measurement & Verification (M&V):
   • Impact: Shifting 1-2 wash cycles/week out of evening peak mitigates 2.0-4.5 kWh of coincident peak demand per home. Across a 1-mill
   meter deployment, this eliminates 150-300 MW of peak ramp capacity, avoiding peaker plant dispatch.
   • M&V Framework: Simple pre/post comparison is confounded by weather and holidays. Utilities must deploy a Randomized Controlled Tr
   (RCT) with Difference-in-Differences (DiD) econometric regression: DDE = (Y_treat,post - Y_treat,pre) - (Y_ctrl,post - Y_ctrl,pre).

2. Model Transferability to Indian Households (REFIT vs. Indian Realities):
   • Cold-Water Machine Architecture: Indian homes predominantly use top-load or semi-automatic machines using cold tap water (or solar
   The power draw is almost purely motor-driven (300-500 W) without the massive 2 kW heating signature, requiring higher SNR detection
   • Grid Voltage Fluctuations: Indian distribution feeders experience wide voltage swings (180 V to 260 V). Active power scales as V^2/R.
   Dynamic voltage normalization (P_norm = P_meas * (V_nom / V_meas)^2) is mandatory before feeding data to disaggregation models.
   • Concurrent Masking Loads: Ceiling fans (70W x 4), evaporative coolers, and inverter/UPS battery charging cycles (300-600 W).

3. Scalability & Impact of 15-Minute vs. 30-Minute Smart Meters:
   • Mathematical Peak Dilution: A 20-min 2,000 W water heating spike is averaged down to a continuous ~667 W plateau over 30 minutes.
   Short motor agitation pulses (150 W) and spin bursts (500 W) are completely filtered out by interval averaging.
   • Pipeline Adaptations: (a) Transition from instantaneous power regression (Watts) to interval energy-block prediction (Wh_interval).
   (b) Train Sequence-to-Point (Seq2Point) convolutional networks with 24-48 timestep receptive fields. (c) Target integer cycle counts
   per ToU tariff block rather than instantaneous curve synthesis, capturing 90% of DSM business value.

Submission Summary: Full modular source code, interactive Jupyter notebooks, generated figures, summary tables, and markdown reports (README.md, DATA.md, RESULTS.md, Recommendation.md) are available in the repository root. Engineered with strict zero-leakage validation and empirical integrity for Flock Energy / Watt Wizards Pvt. Ltd.